

Few-Shot Learning of Tool-Use Skills with Proximity and Tactile Sensing

SONY



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Problem and Motivation

Manipulating **unattached tools**
→ Use tools designed for human
→ Swap tools for different task

Challenge

- **Intrinsic** and **extrinsic** contacts
- **Require many demonstrations** for various tools and environments

Our approach

- 1 **Combine proximity and tactile sensing**
- 2 **Pre-train** feature extraction module in **simulation**
- 3 **Finetune** to **bridge the gap** from simulated to **real target domain**

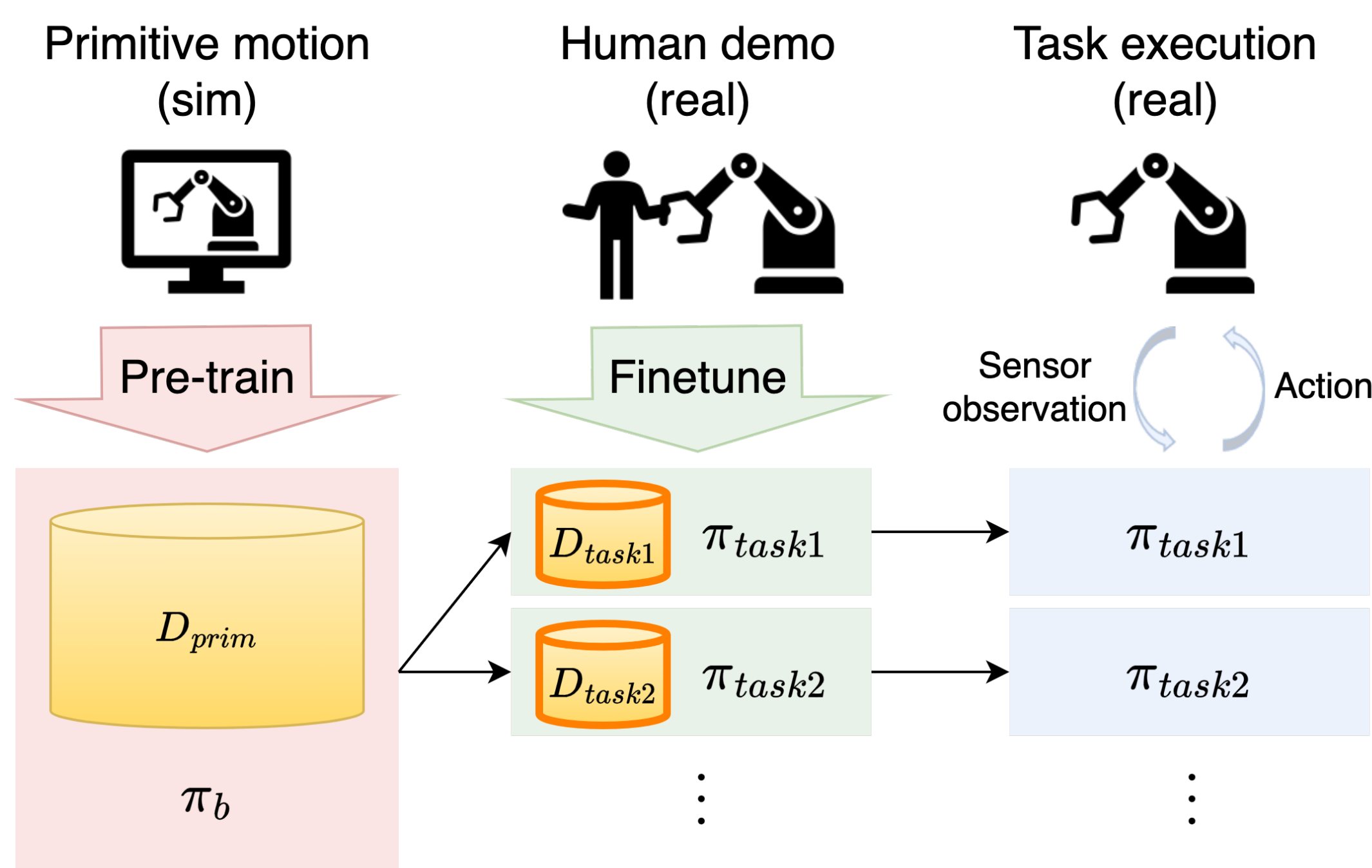
State of the art

- Fixed tool
→ **Intrinsic + extrinsic contacts**
- Single tool
→ **Diverse tools and environments**
- Rigid tools
→ **Deformable/unmodelled tools**

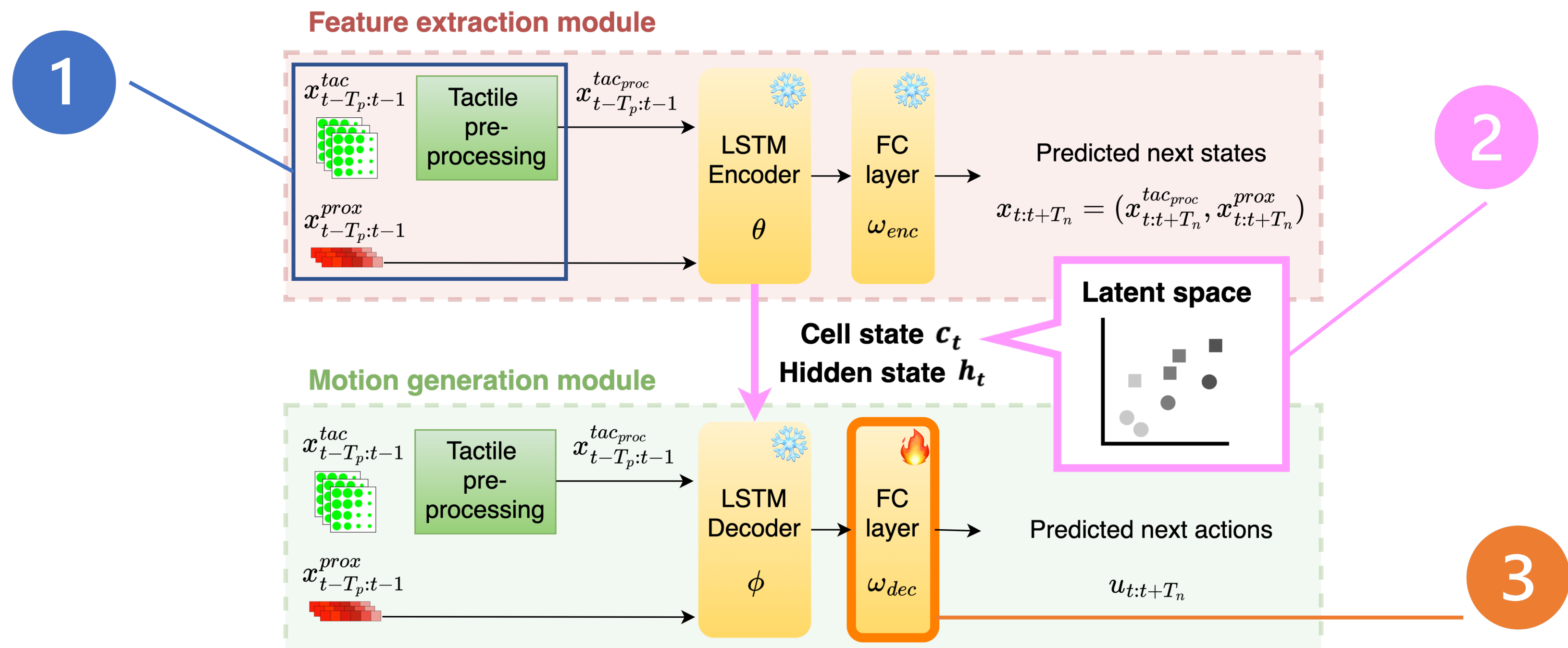
Result

- **Few-shot learning** of tool-use skills
- **Combining proximity and tactile** outperforms single sensor
- **Surface following** tasks using diverse tools

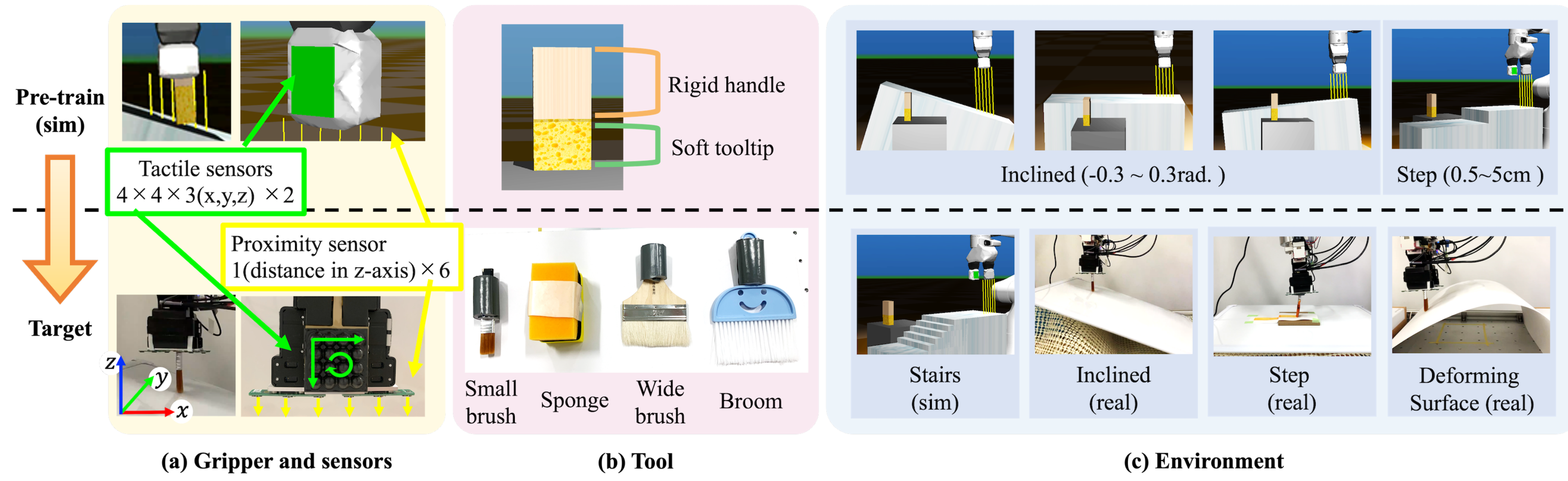
Multimodal few-shot tool-use skill transfer framework



Seq2seq model for π_b and π_{task}



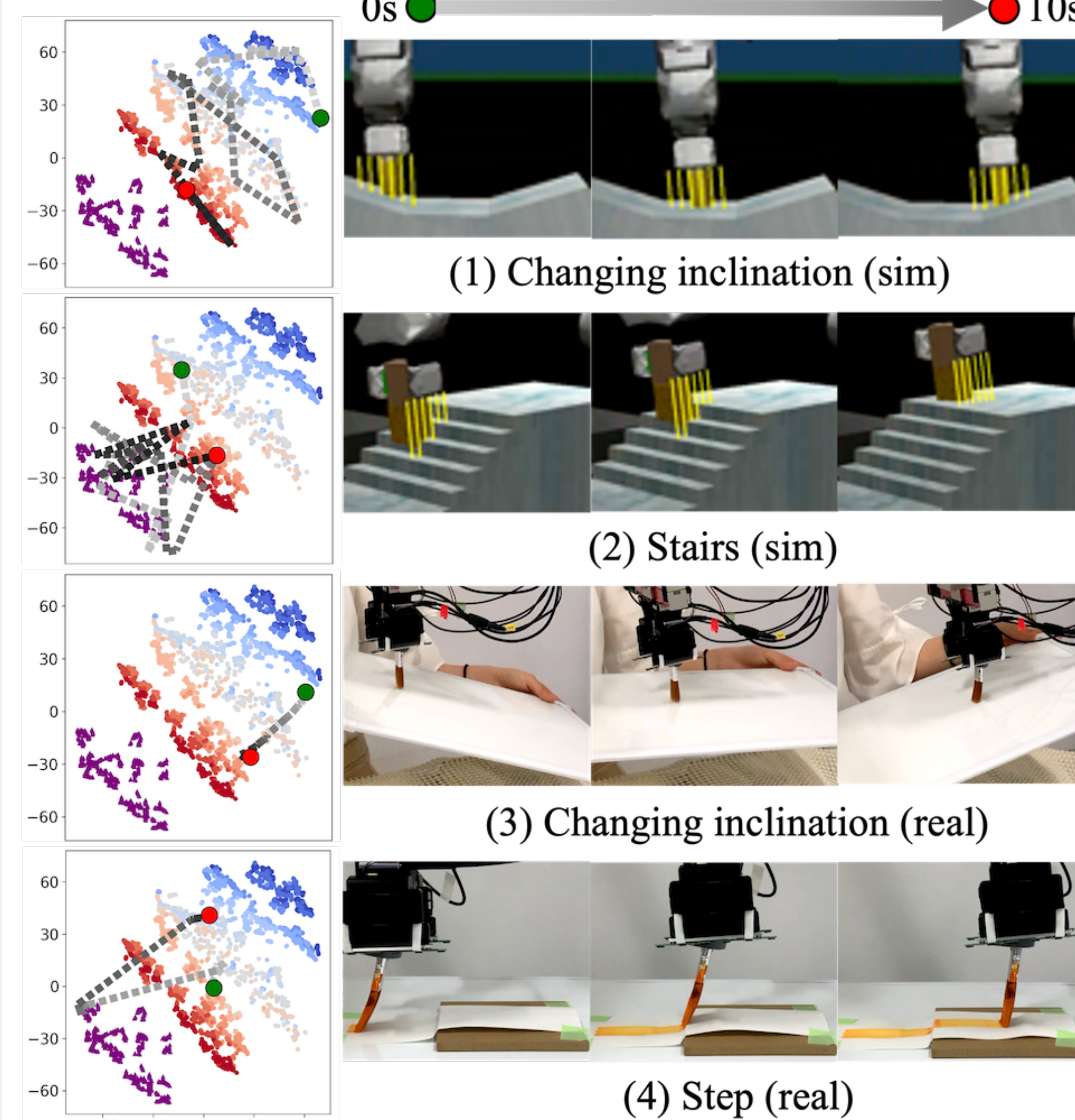
Experimental setup:



Online adaptation:

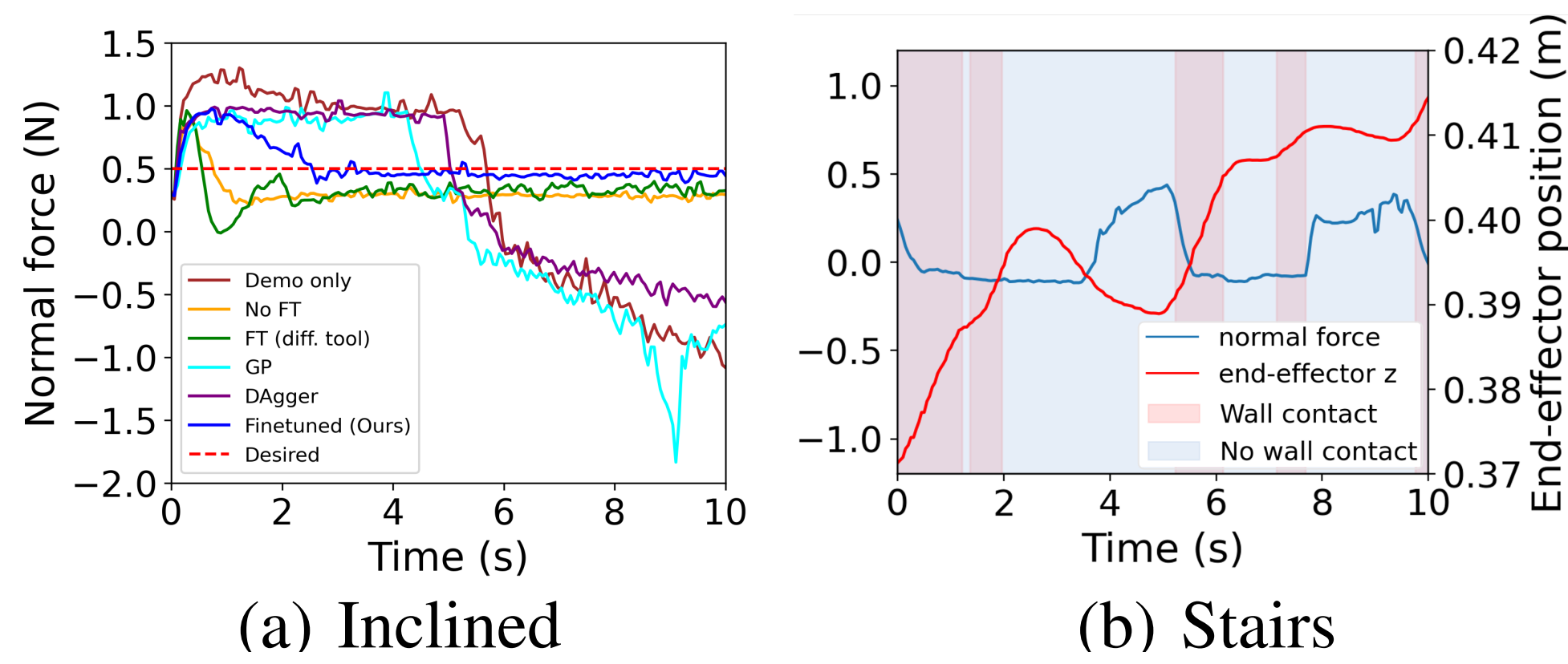
- ✓ Updating latent contact encodings
- ✓ Adapting motions on the fly

Training: inclination: -0.2, -0.1, 0.0, 0.1, 0.2
contact: hitting wall (i.e., $F_x > F_x^{threshold}$ from equation (3))
Test: Start, End, Cell State Evolution



Few-shot tool-use skill performance:

- ✓ Outperforms **Demo-only, No finetuning, and few-shot learning baselines**



Tool	Demo only		No FT		Fintuned	
	x	σ	x	σ	x	σ
Small brush	0.41	0.16	0.25	0.07	0.17	0.02
Sponge	0.30	0.12	0.29	0.01	0.17	0.03
Wide brush	0.18	0.04	0.37	0.12	0.10	0.02
Broom	0.84	0.33	0.66	0.37	0.12	0.05

(a) RMSE of the inclination followed by the end-effector ($\downarrow$).

Tool	Demo only		No FT		Fintuned	
	x	σ	x	σ	x	σ
Small brush	0.041	0.006	0.031	0.011	0.014	0.002
Sponge	0.065	0.034	0.051	0.015	0.017	0.004
Wide brush	0.017	0.008	0.038	0.007	0.009	0.004
Broom	0.034	0.022	0.057	0.047	0.020	0.009

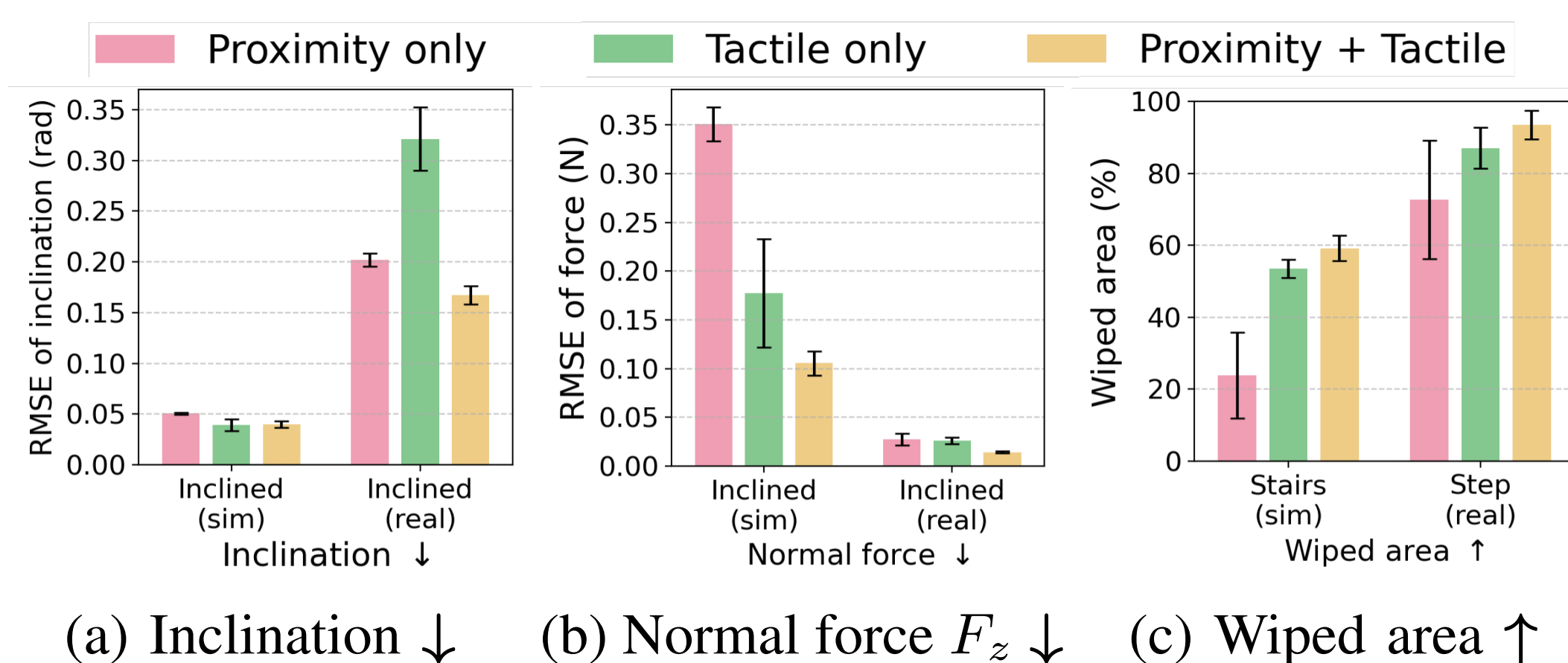
(b) RMSE of the normal contact force ($\downarrow$).

Env.	Demo only		No FT		Fintuned	
	x	σ	x	σ	x	σ
Step	72.74	16.45	87.07	5.61	93.54	4.01

(c) Wiped area ($\uparrow$).

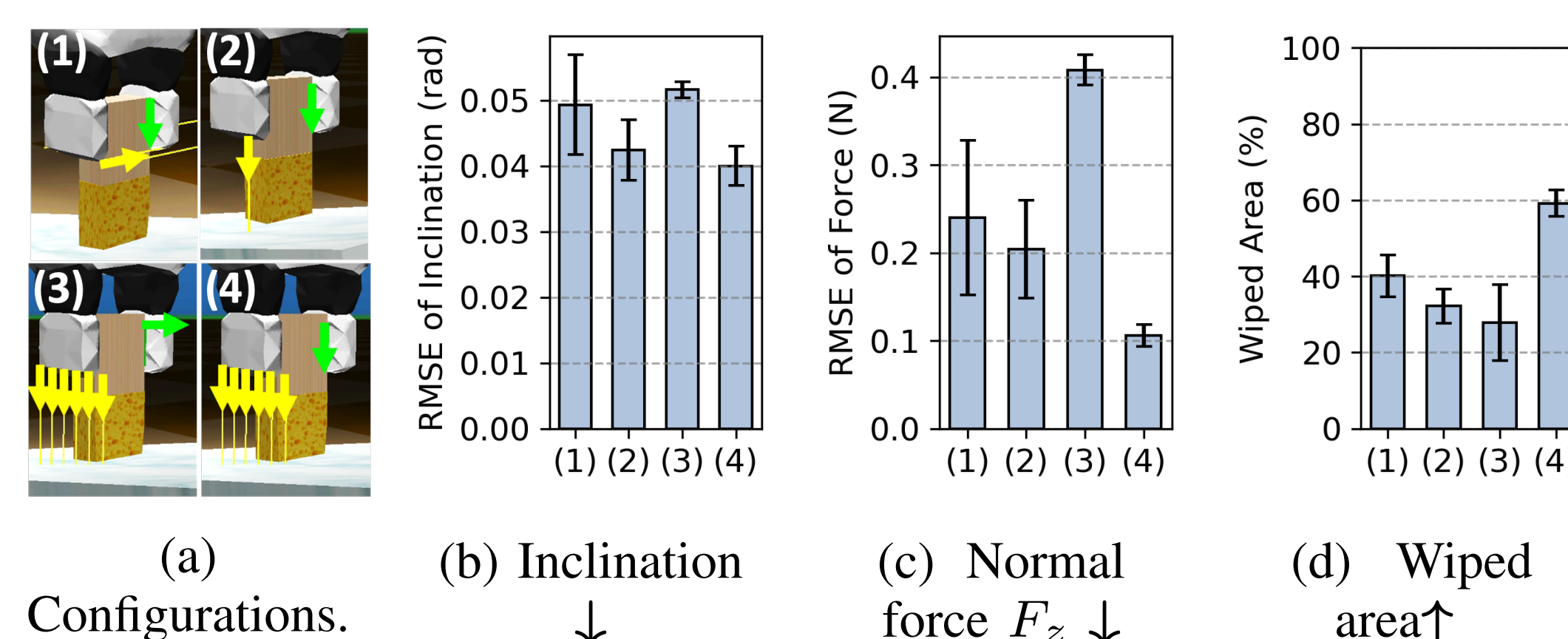
Contribution of each modality:

- ✓ Using both tactile and proximity sensors is essential



Comparison of sensor configurations:

- ✓ Array of proximity sensors and shear force are essential



Project website

